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Autonomous learning of foreign language based on facial emotion recognition and cloud computing

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Abstract. Due to the limitations of the learning environment and unguided guidance, students' autonomous learning of foreign languages after class is not effective. In order to improve the efficiency of autonomous learning of foreign languages, this paper builds a foreign language self-learning system based on facial emotion recognition algorithm and cloud computing platform. Moreover, this paper uses emotion recognition algorithms to identify students' status and guide them to improve students' autonomous learning and improve the system's operating efficiency through cloud computing platforms. In addition, this article combines the needs of autonomous learning to perform facial emotion matching and builds the corresponding functional modules of the system according to the requirements of autonomous learning and designs a sophisticated three-level network structure to achieve a balance between detection performance and real-time performance. In order to verify the performance of the system, an experiment was carried out through the accuracy rate of student's autonomous state emotion recognition and the English improvement of students' autonomous learning. The research results show that the foreign language autonomous learning system constructed in this paper has good performance.

Keywords: Facial emotion, feature recognition, cloud computing, autonomous learning

1. Introduction

As the most widely used language in the world, the role of English on the basis of world economic integration has become increasingly important. How to learn English well has become one of the key points of my country's national quality education.

In the information age, no matter which college, it is necessary to manage a large number of teaching courses and send students, so that students can use their spare time to learn. At the same time, teachers also need to make adjustments to the next teaching curriculum and learning plan according to the teaching situation and student learning situation [1]. The traditional school teaching management model and self-study of students have many drawbacks, such as a wide variety of teaching subjects, complicated management methods, and a lack of systematic management methods. Teaching and students urgently need to use modern Internet technology to complete the business goal of sending teaching content to students' autonomous learning and students' online testing. The traditional method of manual management of teaching materials has become increasingly difficult to adapt to the work of teaching management business in the information age and data age [2].

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With the rapid development of higher education in my country, the investment of universities and colleges in the field of scientific and technological research continues to increase, and new problems continue to arise in the management of teaching and autonomous learning of students. The traditional methods of relying on manual management have deficiencies such as low efficiency, poor timeliness, and standardization. These deficiencies will inevitably hinder the school's teaching work. The proposal of the mobile interactive autonomous learning system has effectively changed the previous inefficient and complicated teaching management model. It solves the management of teachers' teaching work and the orderly management of students' autonomous learning through the computer system, and meets the needs of students to use their spare time to review and learn the knowledge in time. Moreover, it standardizes the daily management of scientific teaching, increases the controllability and efficiency of the teaching work, and improves the students' enthusiasm and learning efficiency. Therefore, it is an effective solution for various schools to improve teaching management standards and student learning levels [3].

With my country's increasing emphasis on education and the continuous improvement of the education system and methods, many places have formed new education models. Some interactive teaching methods are based on multimedia interactive teaching. Students can further learn the knowledge that has not been integrated in the class by watching multimedia courseware in their spare time. At the same time, they can preview the knowledge learned in the future and review the knowledge they have already learned, which effectively improves the efficiency of learning and reduces costs. Some of them use the fragmented time of students' extracurricular time to unify the knowledge taught by the teacher in class, and the students learn through convenient mobile phone viewing methods after class. At the same time, teachers can further deploy students' learning content and homework in the system, and students can also leave a message and discuss for a certain type of problem, and solve the problems encountered in learning in the form of mutual assistance.

2. Related work

Compared with the popularity of the Internet abroad and the maturity of mobile terminals, there is still a big gap between my country's schools and foreign countries in terms of mobile autonomous learning. In the information age, no matter which colleges and universities need to manage a large amount of teaching content and complex business processes. Mobile interactive autonomous learning system is an organic combination of modern computer technology and scientific teaching ideas as well as mobile internet devices. Through the use of computers and peripheral hardware and software devices, it assists students in interactive autonomous learning, promotes teachers' daily management of teaching management and teaching content, and sharing of teaching materials to achieve the informatization of school teaching work [4]. In the context of network information technology, school teaching methods and teaching management also need to keep pace with the times, and its management plan [5] and continuous improvement have become the focus of attention of research institutions in universities and colleges. From abroad, especially in Western countries, the school's teaching management programs and methods are very mature. Students can learn a lot of public lesson resources through the Internet and various mobile terminals in their spare time, which effectively enhances the interest and passion of the whole country. Moreover, the teaching management system of colleges and universities has basically realized automation and informatization [6], such as Cambridge's online learning system for school public courses, and Oxford University's smart mobile phone teaching resource sharing system. Well-known foreign universities such as Harvard and Massachusetts Institute of Technology have also achieved information management and portability of teaching earlier due to advanced research technology and heavy research tasks [7]. There are only a handful of systems specifically aimed at school mobile teaching and school teaching resource sharing. The management of scientific research projects in most colleges and universities is still at the manual stage, and the registration, release, analysis, review, and statistics of teaching information are all carried out under an inefficient management scheme [8].

At present, there is a lack of theoretical research on mobile interactive autonomous learning systems, so it is of great significance to conduct theoretical and practical research on mobile-based interactive learning systems [9].

There are some particularly leading scholars in the field of collaborative filtering technology research [10], and all research is mainly to optimize and improve existing algorithms. The literature [11]

analyzed the traditional collaborative filtering algorithm. In the algorithm, the item similarity is first determined, and two premises are assumed. One is that similar users evaluate different items with similar ratings, and the second is that the same user evaluates similar items with similar ratings. This has led to the emergence of two collaborative filtering algorithms: one is user-based collaborative filtering algorithm, and the other is project-based collaborative filtering algorithm. The literature [12] also analyzed the three similarity calculation methods in the traditional collaborative filtering recommendation algorithm in principle, and pointed out the two major problems, namely data sparseness and cold start. The literature [13] proposed "a collaborative filtering recommendation algorithm that adapts to changes in user interest". In this paper, two weights are optimized: time-based data weights and resource similarity-based data weights. Moreover, it combines the two weights, and finally optimizes the project-based algorithm through the two weights. The literature [14] proposed an algorithm based on item semantic similarity and introduced the concept of "ontology" in philosophy into the algorithm. When calculating similarity, the resources were classified into related fields for calculation. Because the ontology in the field generally has higher similarity, the calculated similarity between resource items has better reliability, and the similarity between users (items) is calculated by the modified cosine similarity formula. When studying traditional algorithms, by characterizing and analyzing the impact of user characteristics and changes in user interest on the algorithm, the literature [15] proposed a "cooperative filtering recommendation algorithm based on user characteristics and user interest changes". This algorithm is more targeted and can better adapt to user changes. The literature [16] taken mobile interactive autonomous learning system as the research object. Starting from the diversity of student learning and the actual learning needs and conditions of students. For many schools, basic campus networks have been built. Gigabit LANs are located in various locations of the school. The school has its own independent server, established a more mature teaching management system and wireless network, and has laboratories in many middle schools and many teacher offices. At the very least, it has been equipped with office computers. With the help of information system construction commonly used in the current informatization construction of all walks of life, it completes an interactive learning platform for students outside of class, and further improves students' independent learning ability. The interactive autonomous learning system makes full use of students' extracurricular time, improves the interaction between students and teachers, and reduces the repetitive investment of teaching funds. Moreover, it helps to promote the country's major strategic policy of cultivating skilled talents and the country's export of skilled talents needed by society [17]. The literature [26] talks about the construction of directed acyclic graph for video coding algorithms for motion estimation in parallel reconfigurable computing systems. The partitioning algorithm also plays a key role in optimizing the encoding of images. The literature [27] dealt with the exploitation of IoT and BigData Analytics using the Hadoop ecosystem in real-time environments. The implementation of IoT-based Smart City is accomplished through the above-mentioned processes. The article [28] centers around IoT and its noteworthy work in sophisticating the human hones and endeavors. This paper moreover overseen the combination of diverse data from distinctive resources that are related with the web. The article [29] talks approximately the different issues within the vehicular communication field with the proposition of agreeable centralized and distributed spectrum detecting model. Due to the execution of the agreeable cognitive model, obstructions and different hidden issues are minimized. The article [30] discusses the problem, such as the tremendous amount of big data, and introduces the SmartBuddy idea of a smart and intelligent world using individual activities and human resources [31, 32].

3. Face detection algorithm based on MTCNN

MTCNN consists of three sub-networks that have been carefully designed, namely P-Net (ProposalNetwork), R-Net (Refine Network), and O-Net (Output Network). Each layer of the three sub-networks has more layers, more parameters, slower speed and higher accuracy than the previous layer. Through the processing of the three sub-networks, the candidate frame coordinates of the face area and the position of the vertical and horizontal coordinates of the facial feature points are finally obtained. In the PRO three-layer sub-network, the relevant predefined parameters are as follows: the convolution kernel step size is 1, the pooling step size is 2, and the number of zero padding layers is 0. Meanwhile, the activation layer linked after the convolution layer

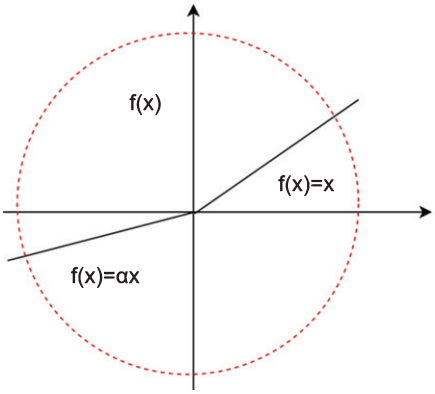

Fig. 1. PReLU response curve.

and the fully connected layer is used PReLU (ParametericRectifiedLiner Unit) as the activation function. The expression of the PReLU activation function is [18]:

$$f(x) = \begin{cases} x, \ if \ x > 0 \\ \alpha x, \ if \ x \leq 0 \end{cases} \tag{1}$$

In the formula, α is the adjustment parameter, and part of the information at $x \leq 0$ is reserved. When $\alpha = 0$, it is equivalent to ReLU activation function. The response curve of PReLU is shown in Fig. 1 [19].

(1) P-Net is the first sub-network of MTCNN. Its main function is to quickly generate candidate windows and candidate box boundary (box regression) vectors. The input of P-Net is an RGB image with a height of 12 pixels wide and a depth of 3 pixels. The input parameters are fixed. The window of 12×12 slides on the input image, and the network needs to determine whether the window of 12×12 contains a face and output the coordinate values of the candidate box of the face area and the face feature points. Its network structure is shown in Fig. 2.

As shown in Fig. 2, the largest pooled kernel size is 3×3. The number of convolution kernels of the second convolutional layer is 16 and the size is 3×3, and the number of convolution kernels of the third convolutional layer is 32 and the size is 3×3. The last layer uses the number of convolution kernels of the third convolutional layer is 32 and the size is 3×3. The last layer uses 1×1-sized convolution kernels for face classification, boundary regression vectors, and face feature point position vectors. P-Net performs Non-Maximum Suppression (NMS) through the obtained boundary coordinates of the face area to merge the bounding boxes of the face area with similar regression vector distance, and the output consists

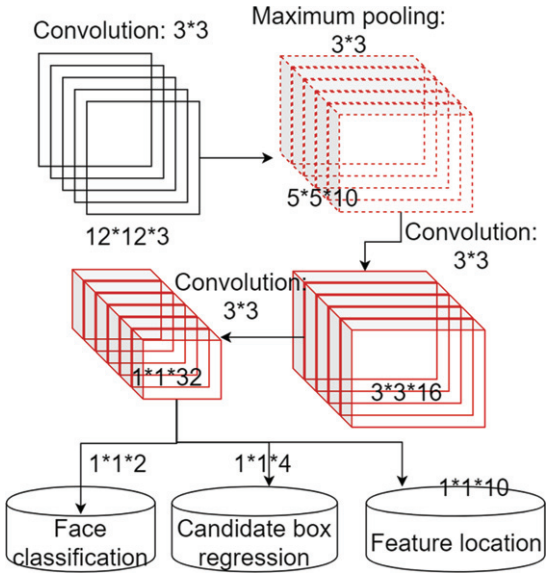

Fig. 2. P-Net network structure.

of three parts 1×1-sized convolution kernels for face classification, boundary regression vectors, and face feature point position vectors. P-Net performs Non-Maximum Suppression (NMS) through the obtained boundary coordinates of the face area to merge the bounding boxes of the face area with similar regression vector distance, and the output consists of three parts [20].

The first part is the face classification. The main function is to determine whether the input window area contains a face. The size of the vector is $1 \times 1 \times 2$, that is, two values, which are the probability of whether the image is a face, and the sum of the two probabilities is equal to 1.

The second part is the candidate box coordinates (bounding box regression), the main function is to output the boundary coordinates of the face area. The size of the vector is $1 \times 1 \times 4$, that is, four values, which are the upper left horizontal coordinate of the boundary, the upper left vertical coordinate, the width of the bounding box, and the height of the bounding box, and the height of the bounding box [21].

The third part is the facial landmark lo-calization. The main function is to output the coordinates of the key points of the face area of the face. The size of the vector is $1 \times 1 \times 10$, that is, ten values, which are the left-eye center coordinates, right-eye center coordinates, nose center coordinates, left mouth corner coordinates, and right mouth corner coordinates. Each coordinate is described by two

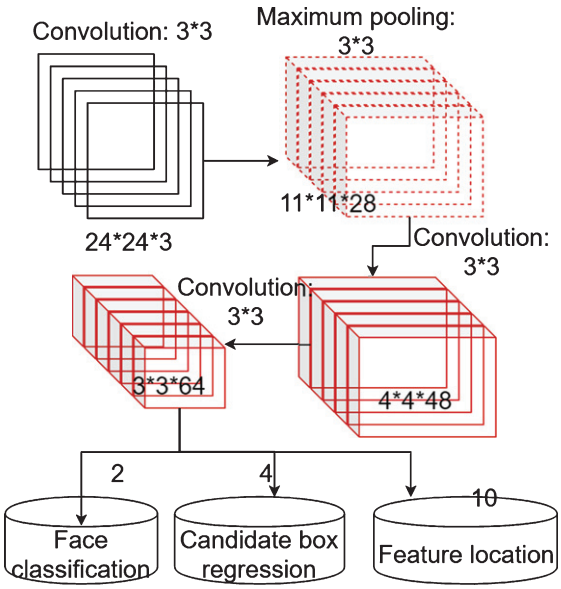

Fig. 3. R-Net network structure.

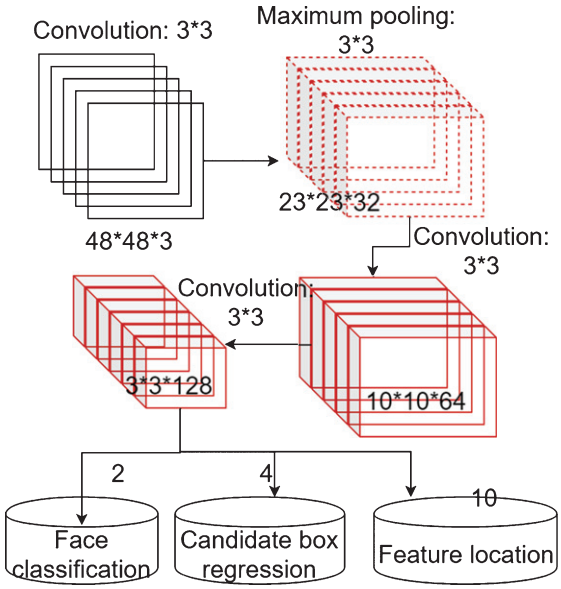

Fig. 4. O-Net network structure.

parameters, abscissa and ordinate, so the output is a 10-dimensional vector.

(2) R-Net is the second sub-network of MTCNN. Its main function is to further judge the areas containing human faces generated by P-Net, delete non-face areas, and reduce the number of calculations and judgments of the next layer of the network. The working principle of R-Net is similar to that of P-Net, and the face area is selected by border regression and non-maximum suppression. However, compared with P-Net, the input size is increased to $24 \times 24 \times 3$, a full link layer is added, and the internal convolution layer is increased. With the increase of parameters, the accuracy rate of face recognition is getting higher and higher, and the face area can be more accurately passed to the next layer of network processing, reducing the calculation of the next layer of network, and the running speed is slower than P-Net. The network structure is shown in Fig. 3 [22].

(3) O-Net is the third sub-network of MTCNN, and its main functions are the same as R-Net. Compared with R-Net, O-Net adds another convolutional layer, and the size of the input image area increases to $48 \times 48 \times 3$. The further increase in the number of parameters leads to a significant increase in accuracy, stronger recognition of the input area, stronger ability to suppress error windows, more refined processing results, fewer face candidates after processing, and higher confidence. The boundary coordinates are more precise. At the same time, in P-Net and O-Net, the accuracy of the coordinates of key points of the face, which is only used to calculate the loss function, is also improved, and the 5-point coordinates are output into a 10-dimensional vector. The network structure is shown in Fig. 4 [23].

The MTCNN network structure can simultaneously handle three tasks: face classification, boundary regression and facial key point extraction. Therefore, during training, MTCNN needs to perform weight learning for three tasks (three loss functions) in a database with three task features (calibration features) simultaneously.

According to Intersection-over-union (IoU), the database is divided into four parts for training according to the ratio of :1 : 1:2, which are negative sample, positive sample, local sample and face key point sample. Among them, positive samples and negative samples are used to train the face region classification task, positive samples and local samples are used to perform the boundary box coordinate prediction training, and face key point samples are used to train the face key point prediction output task [24].

The cross-combination ratio is a standard value used to measure the accuracy of detecting faces in the database. After the MTCNN network, a bounding box is finally output to frame the face area in the image, and the overlap rate between the generated candidate box and the real candidate box marked in the database, that is, that is, the ratio of the intersection and union of the two becomes the measurement

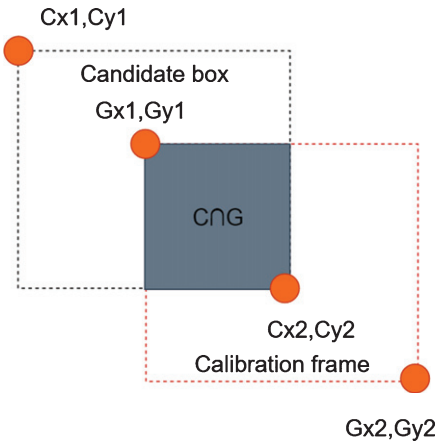

Fig. 5. IOU diagram.

standard for measuring the accuracy of the output. The schematic diagram of IoU is shown in Fig. 5, and the calculation formula is:

$$IOU = \frac{area\,(C) \cap area\,(G)}{area\,(C) \cup area\,(G)} \tag{2}$$

Among them,

$$\begin{aligned} area\,(C) &= (C \times 2 - C \times 1) * (Cy2 - Cy1) \\ area\,(G) &= (G \times 2 - G \times 1) * (Gy2 - Gy1) \end{aligned} \tag{3}$$

is the area of the candidate frame and the calibration frame, respectively.

Negative samples: images with IoU less than 0.3, that is, images that do not contain faces;

Positive samples: images with IoU greater than 0.65, that is, images containing human faces;

Local samples: images with IoU greater than 0.4 and less than 0.65, that is, areas that partially contain human faces;

Samples of key points of the face: images marked with the positions of five key points of the face;

Among them, for images with IoU greater than 0.3 and less than 0.4, due to the calibration error in the database, the boundary between not including faces and some face samples is not clear, it is not suitable for training tasks [25].

The MTCNN network consists of three sub-networks, which are trained separately during training, and each network simultaneously trains multiple different tasks.

(1) Task 1: Face classifier

This part is a simple binary classification task, and the output value represents the probability that the input image is a human face. The training loss function is calculated by using the Cross Entropy function as:

$$L_i^{\det} = -\left(y_i^{\det} \log\left(p_i\right) + \left(1 - y_i^{\det}\right)\left(1 - \log\left(p_i\right)\right)\right) \tag{4}$$

Among them, p_i is the probability that the sample in the input candidate box is a human face after network calculation. $y_i^{\det} \in \{0, 1\}$ is the label value of the sample in the database, 0 represents a non-human face, and 1 represents a human face.

(2) Task 2: Frame coordinate regression

The output of this part is the coordinates of the face candidate frame, including the horizontal coordinate of the upper left corner of the candidate frame, the vertical coordinate of the upper left corner, the width of the candidate frame, and the four-dimensional vector of the height of the candidate frame. The frame coordinate is a regression problem, so the Euclidean distance is used as the learning objective, and the loss function is:

$$L_i^{box} = \left\| \hat{y}_i^{box} - y_i^{box} \right\|_2^2 \tag{5}$$

Among them, $\hat{y}_i^{box}$ is the four-dimensional coordinates of the candidate box output after network calculation, and y_i^{box} is the true coordinate value of the sample in the database.

(3) Task 3: Coordinate positioning of key points on the face

This part of the task is similar to task two, and it is compared with the calibration value in the database. The learning goal is to minimize the error between the network output value and the real value. Here, the Euclidean distance is also used as the learning objective, and the loss function is:

$$L_i^{landamark} = \left\| \hat{y}_i^{landamark} - y_i^{landamark} \right\|_2^2 \tag{6}$$

Among them, $\hat{y}_i^{landamark}$ is the coordinates of the five-point face key points output after network calculation, and $y_i^{landamark}$ is the true coordinate value of the samples in the database. They all contain 10-dimensional vectors, which are the horizontal and vertical coordinates of the left eye, right eye, nose, left corner of the mouth, and right corner of the mouth.

(4) Overall loss function

The three sub-networks have different loss weights for different samples during training according to the different network functions. The overall loss function

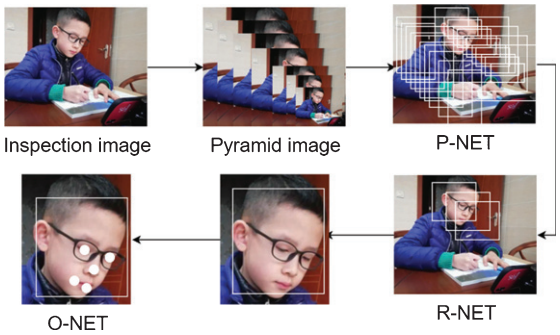

Fig. 6. MTCNN detection flowchart.

after fusion is:

$$L(X_i) = \sum_{i=1}^{N} \sum_{j \in \{dex, box, Iandamark\}} \alpha_j \beta_i^j L_i^j \quad (7)$$

Among them, N is the number of samples used for training, α_j indicates the proportion of the jth sub-network in the overall loss, and β_i^j indicates whether the sample participates in the calculation of j task loss. In P-Net and R-Net networks, the training target focuses on face classification and face frame regression, and the coordinate requirements for key points of the face are not high, so the weight is set to $\alpha_{\text{det}} = 1, \alpha_{box} = 0.5, \alpha_{Iandmark} = 0.5$. In the O-Net network, the training target focuses on outputting the coordinates of key points of the face. In order to obtain higher-precision coordinate values, the weight is set to $\alpha_{\text{det}} = 1, \alpha_{box} = 0.5, \alpha_{Iandmark} = 1$. In each small training batch, the samples are sorted according to the previous item propagation loss, the first 70% are selected as effective samples (hard samples) to calculate the gradient, and other simple samples are ignored.

It is assumed that there is a picture containing a human face, the following uses this picture as an example to illustrate the detection process of the MTCNN algorithm. The flow chart of face detection and alignment by using MTCNN is shown in Fig. 6 below.

4. Face tracking algorithm based on DSST

MOSSE filter: First, a series of image blocks $f_1, f_2, \cdots, f_t$ are extracted from the target as training samples, and the corresponding response values are $g_1, g_2, \cdots, g_t$ after inputting the filter, and the goal of training is to obtain a filter that minimizes the mean square error, that is:

$$\varepsilon = \sum_{j=1}^{t} \left\| h_t * f_j - g_j \right\|^2 = \frac{1}{MN} \left\| H_t F_j - G_j \right\|^2 \quad (8)$$

Through the above formula, the minimum value of ε is found. In the formula, the size of $h_t f_j g_j$ is $M \times N$. The left side of the second equal sign is the equation expression in the time domain, and the right side is the left side expression in the frequency domain, which is converted to each other by Parseval's theorem (Parseval). The solution of the minimum value of ε in the frequency domain is:

$$H_t = \frac{\sum_{j=1}^{t} \bar{G}_j F_j}{\sum_{j=1}^{t} \bar{F}_j F_j} \quad (9)$$

g_j can be any size of output. After the correlation filter is obtained, the image frame Z to be detected is input into the filter, and the correlation score y is obtained:

$$y = F^{-1}(H_t Z) \quad (10)$$

In the formula, F^{-1} represents inverse Fourier transform. When y takes the maximum value, its corresponding position is the updated position of the target of the previous frame in the image frame Z to be detected.

DSST algorithm is improved after adding new scale transformation on the basis of MOSSE algorithm. Based on MOSSE, DSST has designed two independent correlation filter: position filters (translation filter) and size filters (scale filter), to determine the target position and the current scale of the target, respectively.

Different from MOSSE adopting the gray feature of the image block, DSST adopts the feature of histogram of oriented gradient (HOG). From an input image region f, d-dimensional feature vectors are obtained, and a cost function is constructed:

$$\varepsilon = \left\| \sum_{l=1}^{d} h^l * f^l - g \right\| + \lambda \sum_{l=1}^{d} \left\| h^l \right\|^2 \quad (11)$$

In the formula, l represents a certain dimension of the d-dimensional feature vector, and λ is the weight of the regular term. The purpose is to eliminate the zero frequency in the spectrum and prevent the numerator from being zero. In the frequency domain, the optimal correlation filter h that minimizes the cost

function can be obtained:

$$H^l = \frac{\bar{G}F^l}{\sum_{k=1}^{d} \bar{F}^k F^k + \lambda} = \frac{A_t^l}{B_t} \tag{12}$$

In order to speed up the calculation process of the system, the numerator and denominator of the above formula are updated as follows:

$$A_t^l = (1-\eta)\, A_{t-1}^l + \eta \bar{G}_t F_t^l \tag{13}$$

$$B_t = (1-\eta)\, B_{t-1} + \eta \sum_{k=1}^{d} \bar{F}_t^k F_t^k \tag{14}$$

Among them, η is the learning rate. In the newly detected image area, the latest position of the target is the position corresponding to the maximum value output by the relevant filter y:

$$y = F^{-1}\left\{\frac{\sum_{l=1}^{t} \bar{A}^l Z^l}{B+\lambda}\right\} \tag{15}$$

For scale estimation: after obtaining the position coordinates of the tracking target in the current image in the front, based on the center of the candidate position, candidate sizes of different scales are obtained, and then the most matching scale is calculated as the updated value for tracking the current scale. Among them, the selection criteria for scale update are:

$$\alpha^n W \times \alpha^n H, n \in \left\{\left[-\frac{S-1}{2}\right], \cdots, \left[\frac{S-1}{2}\right]\right\} \tag{16}$$

In the formula, W and H are the scale size of the last tracking. $\alpha = 1.02$ is the scaling factor, $S = 33$ is the number of candidate scales, and the matching direction is from small to large.

The schematic diagram of the DSST algorithm is shown in Fig. 7.

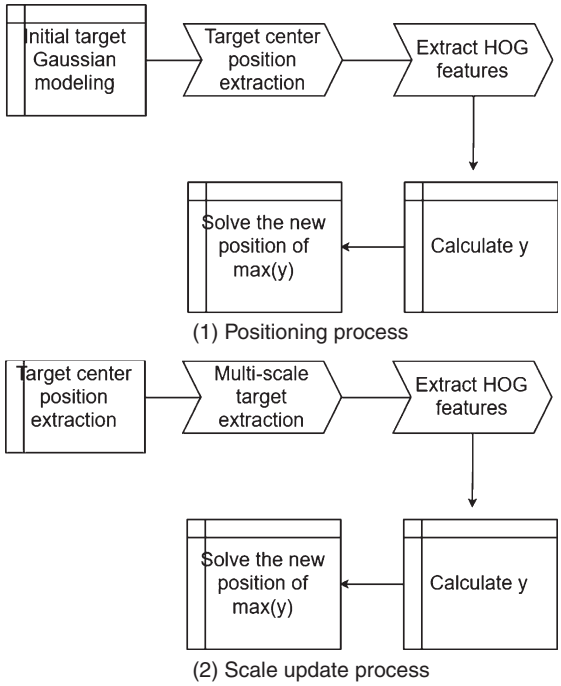

Fig. 7. DSST algorithm flowchart.

5. Face feature point detection based on regression tree set

The ERT feature point extraction algorithm is a regression-based method. The core content is the use of two-layer cascade.

Before describing the ERT model, the expression of facial feature points in the image is described first. It is assumed that $x_i \in \square^2$ is the 2-dimensional coordinate value in the x, y coordinate system of the i-th feature in the facial feature points of the image.

Vector

$$S = \left(x_1^T, x_2^T, \cdots, x_P^T\right) \in X^{2P} \tag{17}$$

is a vector composed of the coordinates of P facial feature points in the image, and is also called the shape vector of image I.

The first layer adopts the method of cascade regression to establish the ERT model. Its iteration formula is:

$$\hat{S}^{t+1} = \hat{S}^{(t)} + \gamma_t\left(I, \hat{S}^{(t)}\right) \tag{18}$$

$$\Delta S_i^{(t)} = S_{\pi_i} - \hat{S}_i^{(t)} \tag{19}$$

In the formula, $\hat{S}^{(t)}$ is the shape vector, including the coordinates of all the feature points of the face. γ_t is the first-level regressor, the input is the training image I and the current shape vector, and the output generates the shape vector update value for this iteration. $\Delta S_i^{(t)}$ is the difference between the output of the regressor and the true value. In the training of γ_t, the gradient tree boosting algorithm based on the sum of square error as the loss function is used. After iteration, the updated values of the shape vectors of the facial feature points can be obtained. The updated values generated by the iteration are used to update the positions of all the key points at once, so

that the error between the estimated position of all key points and the true position of the key points is minimized, and the positioning accuracy of the key points is improved.

It is assumed that there is a training set $(I_1, S_1), (I_2, S_2), \cdots, (I_n, S_n)$, among them, I_i represents the face picture and S_i represents the shape vector of the facial feature points in picture I_i. The iteration formula is:

$$\pi_i \in \{1, \cdots, n\} \tag{20}$$

$$\hat{S}_i^{(0)} \in \{S_1, S_2, \cdots, S_n\} \setminus S_{\pi_i} \tag{21}$$

$$\Delta S_i^{(0)} = S_{\pi_i} - \hat{S}_i^{(0)} \tag{22}$$

$\hat{S}_i^{(0)}$ is the average feature point vector of the shape vector S in the training picture calculated according to Equations (20), (21), and (22), which is used as the initial shape vector for cyclic iterative update. n represents the number of training pictures, and $\Delta S_i^{(0)}$ represents the difference between the initial value of the shape vector of the real mark of the input picture and the shape vector. Through the above iteration method, the iteration is continuously carried out. When the number of layers of the first-level regression cascade is k layers, k regressors $\gamma_1, \gamma_2, \cdots, \gamma_k$ are generated after multiple iterations, and these regressors are the regression models we want to get.

The second layer is mainly the training of the regressor γ_i, which mainly uses the gradient lifting tree method for training. The training algorithm is as follows:

(1) The training data set is set to $\left\{I_{\pi_i}, \hat{S}_i^{(t)}, \Delta S_i^{(t)}\right\}_{i=1}^{N}$ and the learning rate is $0 < \nu < 1$

(2) Initialization:

$$f_0\left(I, \hat{S}^{(t)}\right) = \underset{\gamma \in \square^{2p}}{\arg\min} \sum_{i=1}^{N} \left\| \Delta S_i^{(t)} - \gamma \right\|^2 \tag{23}$$

(3) The value of r_{ik} is updated every iteration:

$$r_{ik} = \Delta S_i^{(t)} - f_{k-1}\left(I_{\pi_i}, \hat{S}_i^{(t)}\right) \tag{24}$$

(4) The weak regression equation $g_k\left(I, \hat{S}^{(t)}\right)$ is obtained by r_{ik}, and then iterates to update $f_k\left(I, \hat{S}^{(t)}\right)$.

$$f_k\left(I, \hat{S}^{(t)}\right) = f_{k-1}\left(I, \hat{S}^{(t)}\right) + \nu g_k\left(I, \hat{S}^{(t)}\right) \tag{25}$$

(5) The algorithm loops and iterates in step (3)-step (4) until it converges or reaches the set number of iterations, the final output is:

$$r_t\left(I, \hat{S}^{(t)}\right) = f_k\left(I, \hat{S}^{(t)}\right) \tag{26}$$

It is particularly important to note that in each two-level regression, the object input to the regressor is $\Delta S_i^{(t)}$ after the completion of each first-level regression, that is, the difference between the predicted value and the true value. In the second-level regression, the squared error is used:

$$\varepsilon = \frac{1}{2}\left[y_i - f(x_i)\right]^2 \tag{27}$$

As a loss function, $y_i - f(x_i)$ is obtained after derivation, and this gradient is used as the fitting object in each round of the iteration to obtain the final construction model.

The following will briefly introduce the search and fitting process of the model, the process is as follows:

(1) From the face candidate region of the input point, facial feature points are detected, and the coordinate values of the feature points are initialized as shape vectors in the form of $S = \left(x_1^T, x_2^T, \cdots, x_P^T\right) \in \square^{2P}$;

(2) The pixel difference feature between all facial feature point pairs is calculated to split the tree;

(3) All decision trees are used to find residuals. By comparing the thresholds saved by non-leaf nodes with the pixel difference values, the residuals saved by the leaf nodes of each tree can be obtained. The residuals of all subtrees are summed to obtain the global residual;

(4) The global residual is added to the result of the previous iteration;

(5) Processes (2)–(4) are repeated until the iteration is completed.

It can be found that: When $T = 0$, the face shape vector is the initial value. After the first secondary cascade regression, the error between the facial feature point and the actual position has been significantly reduced. As the number of cascading regression trees increases, the estimated coordinate values are getting closer to the true coordinate values. when $T = 10$, the error of the ERT fitting feature point shape vector coordinate and the training set marked coordinate point is extremely small, indicating that the algorithm has excellent performance.

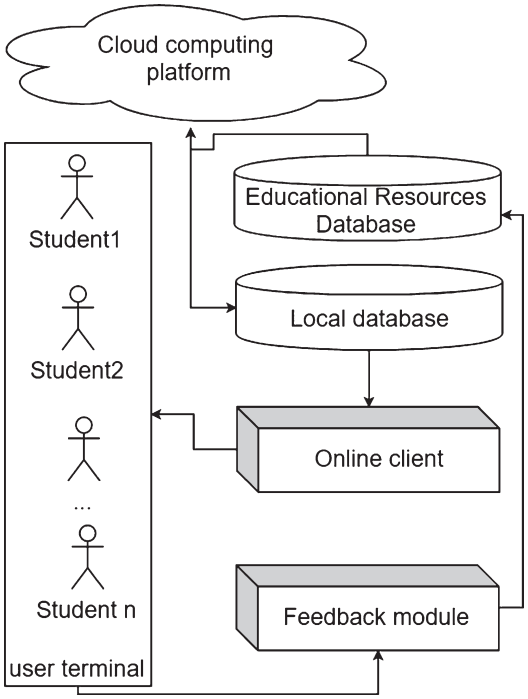

Fig. 8. System structure model.

Table 1
Statistical table of emotion recognition results

	1	2	3	4	5	6
Listen carefully	86.22%	85.13%	88.79%	92.59%	87.36%	86.89%
Daze	89.68%	86.70%	92.04%	86.26%	91.86%	94.76%
Patient	90.26%	88.61%	92.45%	94.52%	93.33%	91.75%
Sleepy	93.41%	86.87%	94.41%	92.40%	87.47%	90.26%
Thinking	94.36%	92.30%	90.21%	85.42%	85.31%	91.08%
Semi-listening	89.36%	90.90%	87.98%	89.34%	92.69%	88.89%

6. Model performance test

Based on the above analysis, first, a cloud computing foreign language autonomous learning model based on facial emotion recognition is constructed. The model structure is shown in Fig. 8.

The state recognition system in this paper is fed back to the cloud through the feedback module, and then the students' facial emotions are recognized by the algorithm in this paper. In order to verify the performance of the algorithm in this paper, we first study the accuracy of sentiment verification by the algorithm. The study uses several states of careful listening, daze, patience, sleepiness, thinking, and semi-listening to judge emotions and judge the recognition rate of these groups of emotions. In this study, a total of 6 groups of tests are carried out. The results obtained are shown in Table 1 and Fig. 9.

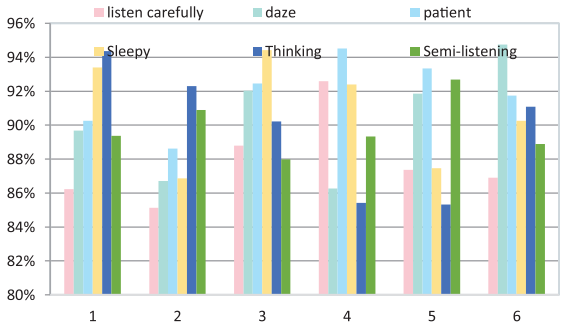

Fig. 9. Statistical diagram of emotion recognition results.

Table 2
Statistical table of foreign language autonomous learning performance

	Before testing	After testing
1	80	93
2	78	83
3	78	83
4	76	82
5	71	81
6	70	80
7	69	80
8	68	77
9	67	75
10	65	74
11	64	73
12	58	73
13	57	71
14	55	71
15	49	68
16	47	67
17	47	63
18	39	59
19	37	58
20	35	57

As shown in Fig. 9, the system constructed in this paper has the worst recognition effect in the cognitive listening part. The reason is that this part is more similar to a daze expression, and it is prone to confusion. However, the minimum recognition rate is also above 85%, so it can meet the actual needs. After that, the effect of the system's English autonomous learning is studied and analyzed. A total of 20 students are analyzed and their English scores are compared before and after the test. The results are shown in Table 2 and Fig. 10.

As shown in the radar chart shown in Fig. 10, the red area indicating the English academic performance after the test surrounds the green portion indicating

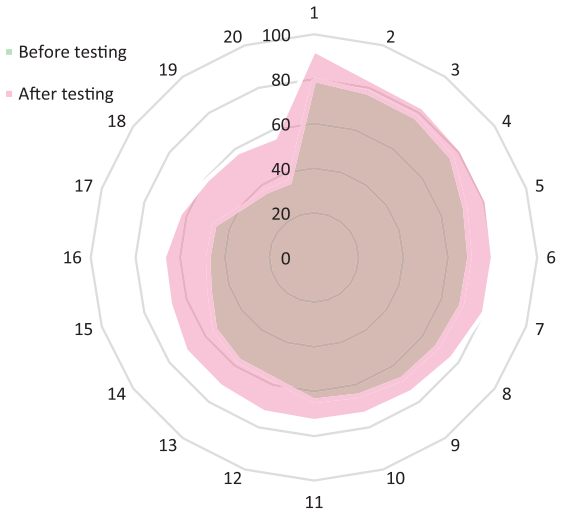

Fig. 10. Statistical diagram of foreign language autonomous learning performance.

the English academic performance before the test, that is, everyone's English is improved to a certain extent. From this we can see that the system proposed in this paper works well.

7. Conclusion

The operation of a foreign language autonomous learning system based on facial emotion recognition and cloud computing has injected advanced management technology and a scientific and reasonable management system into the effective management and arrangement of school teaching. Moreover, it enables students to study autonomously in their spare time and make full use of the student's fragmented time, which improves students' enthusiasm and learning efficiency. In addition, through the integration of curriculum standards, management methods, note exchange and unit testing, students' learning interest and achievements are improved, which ensures the school's purpose of improving teaching efficiency. The system provides a convenient interactive learning platform, so that students can learn knowledge more freely and openly and share the learned knowledge with other users. At the same time, the system also sorts the shared data, so that students can get better learning experience of others in the first time, thereby effectively improving their own learning efficiency. Finally, the system also provides a mandatory testing mechanism. Only after completing each unit test can the course test be carried out, which makes the management mechanism more complete and enables users to learn more targeted.

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